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Harness engineering: leveraging Codex in an agent-first world

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Over the past five months, our team has been running an experiment: building and shipping an internal beta of a software product with **0 lines of manually-written code**.

The product has internal daily users and external alpha testers. It ships, deploys, breaks, and gets fixed. What's different is that every line of code—application logic, tests, CI configuration, documentation, observability, and internal tooling—has been written by Codex. We estimate that we built this in about 1/10th the time it would have taken to write the code by hand.

Humans steer. Agents execute.

We intentionally chose this constraint so we would build what was necessary to increase engineering velocity by orders of magnitude. We had weeks to ship what ended up being a million lines of code. To do that, we needed to understand what changes when a software engineering team's primary job is no longer to write code, but to design environments, specify intent, and build feedback loops that allow Codex agents to do reliable work.

This post is about what we learned by building a brand new product with a team of agents—what broke, what compounded, and how to agents how to work in the repository was itself written by Codex.

There was no pre-existing human-written code to anchor the system. From the beginning, the repository was shaped by the agent.

Five months later, the repository contains on the order of a million lines of code across application logic, infrastructure, tooling, documentation, and internal developer utilities. Over that period, roughly 1,500 pull requests have been opened and merged with a small team of just three engineers driving Codex. This translates to an average throughput of 3.5 PRs per engineer per day, and surprisingly the throughput has *increased* as the team has grown to now seven engineers. Importantly, this wasn't output for output's sake: the product has been used by hundreds of users internally, including daily internal power users.

Throughout the development process, humans never directly contributed any code. This became a core philosophy for the team: **no manually-written code**.

Redefining the role of the engineer

The lack of hands-on human coding **introduced a different kind of engineering work, focused on systems, scaffolding, and leverage**.

Early progress was slower than we expected, not because Codex was incapable, but because the environment was underspecified. The agent lacked the tools, abstractions, and internal structure required to make progress toward high-level goals. The primary job of our engineering team became enabling the agents to do useful work.

In practice, this meant working depth-first: breaking down larger goals into smaller building blocks (design, code, review, test, etc), prompting the agent to construct those blocks, and using them to unlock more complex tasks. When something failed, the fix was almost never "try harder." Because the only way to make progress was to get Codex to do the work, human engineers always stepped into the task and asked: "what capability is missing, and how do we make it both legible and enforceable for the agent?"

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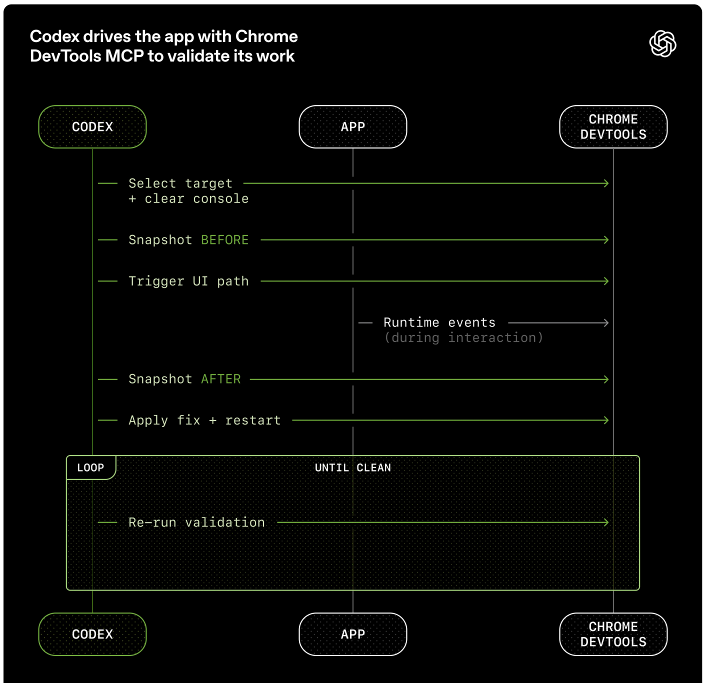
Humans may review pull requests, but aren't required to. Over time, we've pushed almost all review effort towards being handled agent-to-agent.

Increasing application legibility

As code throughput increased, our bottleneck became human QA capacity. Because the fixed constraint has been human time and attention, we've worked to add more capabilities to the agent by making things like the application UI, logs, and app metrics themselves directly legible to Codex.

For example, we made the app bootable per git worktree, so Codex could launch and drive one instance per change. We also wired the Chrome DevTools Protocol into the agent runtime and created skills for working with DOM snapshots, screenshots, and navigation. This enabled Codex to reproduce bugs, validate fixes, and reason about UI behavior directly.



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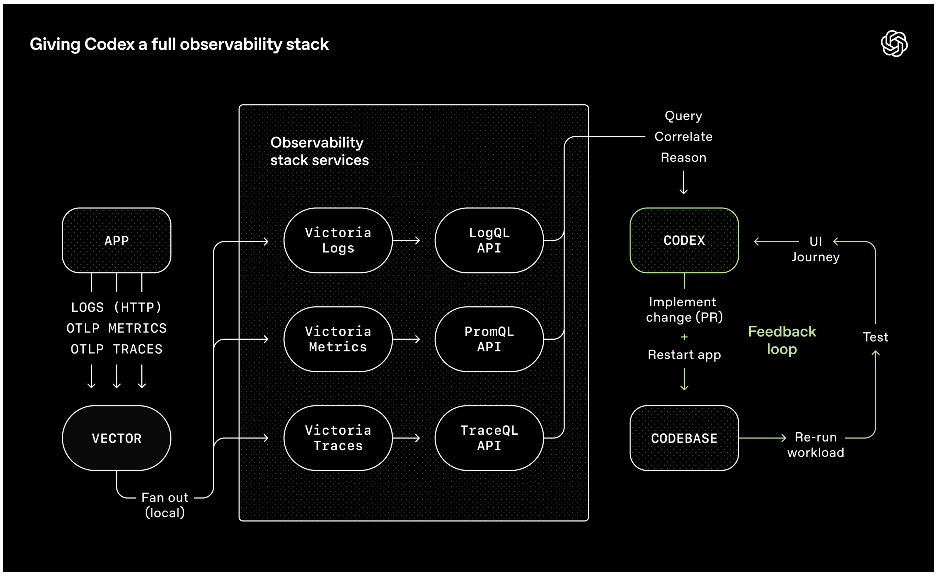

We regularly see single Codex runs work on a single task for upwards of six hours (often while the humans are sleeping).

We made repository knowledge the system of record

Context management is one of the biggest challenges in making agents effective at large and complex tasks. One of the earliest lessons we learned was simple: **give Codex a map, not a 1,000-page instruction manual.**

We tried the "one big `AGENTS.md`" approach. It failed in predictable ways:

- **Context is a scarce resource.** A giant instruction file crowds out the task, the code, and the relevant docs—so the agent either misses key constraints or starts optimizing for the wrong ones.
- **Too much guidance becomes *non-guidance*.** When everything

the table of contents.

The repository's knowledge base lives in a structured `docs/` directory treated as the system of record. A short `AGENTS.md` (roughly 100 lines) is injected into context and serves primarily as a map, with pointers to deeper sources of truth elsewhere.

Plain Text

```
AGENTS.md
ARCHITECTURE.md
docs/
├── design-docs/
│   ├── index.md
│   ├── core-beliefs.md
│   └── ...
├── exec-plans/
│   ├── active/
│   ├── completed/
│   └── tech-debt-tracker.md
├── generated/
│   └── db-schema.md
├── product-specs/
│   ├── index.md
│   ├── new-user-onboarding.md
│   └── ...
├── references/
│   ├── design-system-reference-llms.txt
│   ├── nixpacks-llms.txt
│   ├── uv-llms.txt
│   └── ...
├── DESIGN.md
├── FRONTEND.md
├── PLANS.md
├── PRODUCT_SENSE.md
└── QUALITY_SCORE.md
```

In-repository knowledge store layout.

Design documentation is catalogued and indexed, including verification status and a set of core beliefs that define agent-first operating principles. Architecture documentation provides a top-level map of domains and package layering. A quality document grades each product domain and architectural layer, tracking gaps over time.

Plans are treated as first-class artifacts. Ephemeral lightweight plans

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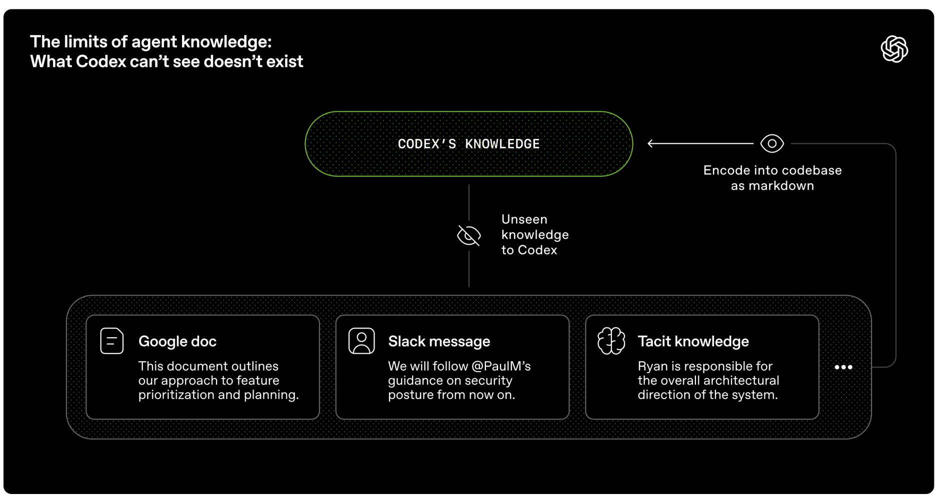
that the knowledge base is up to date, cross-linked, and structured correctly. A recurring “doc-gardening” agent scans for stale or obsolete documentation that does not reflect the real code behavior and opens fix-up pull requests.

Agent legibility is the goal

As the codebase evolved, Codex’s framework for design decisions needed to evolve, too.

Because the repository is entirely agent-generated, it’s optimized first for *Codex’s legibility*. In the same way teams aim to improve navigability of their code for new engineering hires, our human engineers’ goal was making it possible for an agent to reason about the full business domain **directly from the repository itself.**

From the agent’s point of view, anything it can’t access in-context while running effectively doesn’t exist. Knowledge that lives in Google Docs, chat threads, or people’s heads are not accessible to the system. Repository-local, versioned artifacts (e.g., code, markdown, schemas, executable plans) are all it can see.



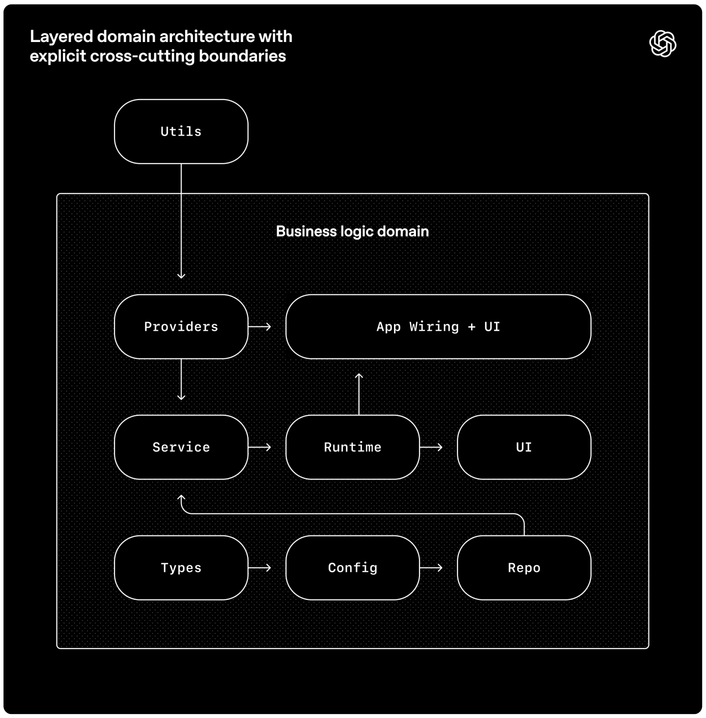
This framing clarified many tradeoffs. We favored dependencies and abstractions that could be fully internalized and reasoned about in-repo. Technologies often described as “boring” tend to be easier for agents to model due to composability, api stability, and

Increasing application legibility

Documentation alone doesn’t keep a fully agent-generated codebase coherent. **By enforcing invariants, not micromanaging implementations, we let agents ship fast without undermining the foundation.** For example, we require Codex to parse data shapes at the boundary, but are not prescriptive on how that happens (the model seems to like Zod, but we didn’t specify that specific library).

Agents are most effective in environments with strict boundaries and predictable structure, so we built the application around a rigid architectural model. Each business domain is divided into a fixed set of layers, with strictly validated dependency directions and a limited set of permissible edges. These constraints are enforced mechanically via custom linters (Codex-generated, of course!) and structural tests.

The diagram below shows the rule: within each business domain (e.g. App Settings), code can only depend “forward” through a fixed set of layers (Types → Config → Repo → Service → Runtime → UI). Cross-cutting concerns (auth, connectors, telemetry, feature flags) enter through a single explicit interface: Providers. Anything else is



This is the kind of architecture you usually postpone until you have hundreds of engineers. With coding agents, it's an early prerequisite: the constraints are what allows speed without decay or architectural drift.

In practice, we enforce these rules with custom linters and structural tests, plus a small set of "taste invariants." For example, we statically enforce structured logging, naming conventions for schemas and types, file size limits, and platform-specific reliability requirements with custom lints. Because the lints are custom, we write the error messages to inject remediation instructions into agent context.

In a human-first workflow, these rules might feel pedantic or constraining. With agents, they become multipliers: once encoded, they apply everywhere at once.

At the same time, we're explicit about where constraints matter and where they do not. This resembles leading a large engineering platform organization: enforce boundaries centrally, allow autonomy comments, refactoring pull requests, and user-facing bugs are captured as documentation updates or encoded directly into tooling. When documentation falls short, we promote the rule into code

Throughput changes the merge philosophy

As Codex's throughput increased, many conventional engineering norms became counterproductive.

The repository operates with minimal blocking merge gates. Pull requests are short-lived. Test flakes are often addressed with follow-up runs rather than blocking progress indefinitely. In a system where agent throughput far exceeds human attention, corrections are cheap, and waiting is expensive.

This would be irresponsible in a low-throughput environment. Here, it's often the right tradeoff.

What "agent-generated" actually means

When we say the codebase is generated by Codex agents, we mean everything in the codebase.

Agents produce:

- Product code and tests
- CI configuration and release tooling
- Internal developer tools
- Documentation and design history
- Evaluation harnesses
- Review comments and responses
- Scripts that manage the repository itself
- Production dashboard definition files

Humans always remain in the loop, but work at a different layer of

Increasing levels of autonomy

As more of the development loop was encoded directly into the system—testing, validation, review, feedback handling, and recovery—the repository recently crossed a meaningful threshold where Codex can end-to-end drive a new feature.

Given a single prompt, the agent can now:

- Validate the current state of the codebase
- Reproduce a reported bug
- Record a video demonstrating the failure
- Implement a fix
- Validate the fix by driving the application
- Record a second video demonstrating the resolution
- Open a pull request
- Respond to agent and human feedback
- Detect and remediate build failures
- Escalate to a human only when judgment is required
- Merge the change

This behavior depends heavily on the specific structure and tooling of this repository and should not be assumed to generalize without similar investment—at least, not yet.

Entropy and garbage collection

Full agent autonomy also introduces novel problems. Codex replicates patterns that already exist in the repository—even uneven or suboptimal ones. Over time, this inevitably leads to drift.

Initially, humans addressed this manually. Our team used to spend every Friday (20% of the week) cleaning up “AI slop.” Unsurprisingly, that didn’t scale.

Instead, we started encoding what we call “golden principles” directly interest loan: it’s almost always better to pay it down continuously in small increments than to let it compound and tackle it in painful bursts. Human taste is captured once, then enforced continuously on every line of code. This also lets us catch and resolve bad patterns on a daily basis, rather than letting them spread in the code base for days or weeks.

What we’re still learning

This strategy has so far worked well up through internal launch and adoption at OpenAI. Building a real product for real users helped anchor our investments in reality and guide us towards long-term maintainability.

What we don’t yet know is how architectural coherence evolves over years in a fully agent-generated system. We’re still learning where human judgment adds the most leverage and how to encode that judgment so it compounds. We also don’t know how this system will evolve as models continue to become more capable over time.

What’s become clear: building software still demands discipline, but the discipline shows up more in the scaffolding rather than the code. The tooling, abstractions, and feedback loops that keep the codebase coherent are increasingly important.

Our most difficult challenges now center on designing environments, feedback loops, and control systems that help agents accomplish our goal: build and maintain complex, reliable software at scale.

As agents like Codex take on larger portions of the software lifecycle, these questions will matter even more. We hope that sharing some early lessons helps you reason about where to invest your effort so you can just build things.

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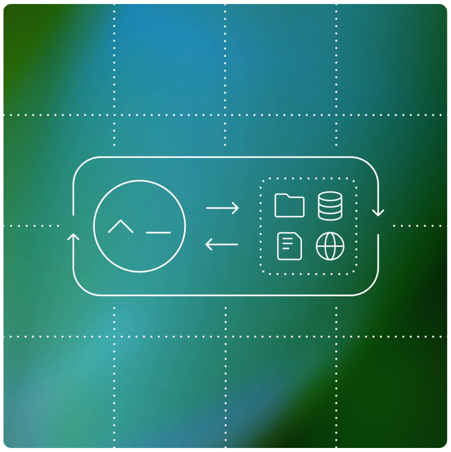

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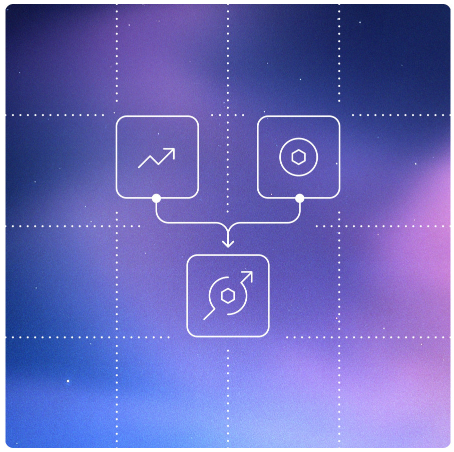

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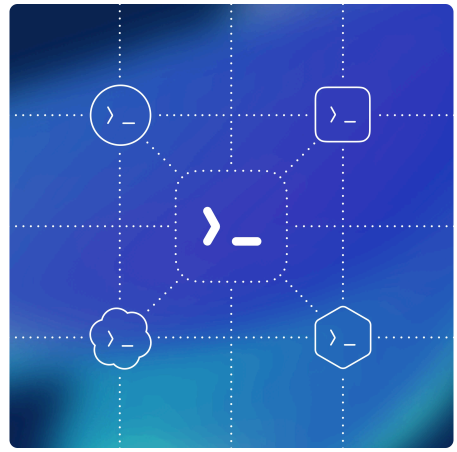

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